

Christopher Musco

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Education

Massachusetts Institute of Technology

Ph.D. Candidate, Computer Science – Theory of Computation

Advisor: Jonathan Kelner

Masters Thesis: Dimensionality Reduction for Sparse and Structured Matrices (Spring 2015)

Cambridge, MA

2013 – present

Yale University

B.S. Applied Mathematics, B.S. Computer Science

Applied Mathematics Thesis: Fast Discrete Laplace Transforms (with Vladimir Rokhlin)

Computer Science Thesis: Graph Constructions for Machine Learning (with Daniel Spielman)

New Haven, CT

2008 – 2012

Research Areas

theory of algorithms • foundations of data science • scalable machine learning • randomized linear algebra
dimensionality reduction • sketching and streaming • algorithmic graph theory • iterative matrix algorithms

Publications

Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science.

Conference and Journal Papers.....

Stability of the Lanczos Method for Matrix Function Approximation. Cameron Musco, Christopher Musco, Aaron Sidford. *SODA* 2018.

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *NIPS* 2017.

Random Fourier Features for Kernel Ridge Regression. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ICML* 2017.

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *SODA* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *ALLENEX* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *ICML* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *NIPS* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *STOC* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiroglou-Douskos, Martin Rinard. *NDSS* 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *ITCS* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco,

Christopher Musco, Aaron Sidford. *FOCS* 2014, *SIAM Journal on Computing* 2017.

Teaching and Mentorship

Advanced Algorithms (MIT 6.854/18.415)

Teaching Assistant

2016

Held weekly office hours, wrote problem sets, and compiled notes and resources for an updated version of this capstone graduate algorithms course. Wrote guide on topics and expectations for student projects. Prepared and delivered a lecture on linear programming relaxations.

Technical Communication Skills for Graduate Students (MIT 6.S977)

Workshop Leader

2016

Lead weekly workshops that supplemented lecture material with hands-on writing and presentation exercises. Met with students one-on-one outside of class to offer individual support and coaching in technical communication.

MIT Graduate Communication Lab

Communication Advisor

2015 – present

Helping to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offer feedback, guidance, and coaching to individual students during weekly office hours (75+ individual appointments to date).

Research Science Institute at MIT

Summer Research Advisor

2014

Mentored Christopher Wang, a high school student participating in the prestigious RSI summer program. Chris' project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Introduction to Computer Science (Yale CPSC-201)

Teaching Assistant

2012

Covered functional programming, boolean algebra, formal language theory, and basic complexity theory.

Introduction to Programming (Yale CPSC-112)

Teaching Assistant

2011

Led weekly section for 15-25 students covering basic Java programming. Tutored 6 students individually.

Introductory Calculus (Yale MATH-112 & MATH-115)

Grader

2009 – 2010

Scored problem sets and provided explanations and solutions for students, as well as feedback to the professor.

Industry Research

Yahoo Labs

New York, NY

Research Intern, Scalable Machine Learning Group

Summer 2015

Mentors: Justin Thaler, Maxim Srividenko, Edo Liberty

Worked on theoretical matrix sampling problems and on algorithms for managing Yahoo's Fantasy Sports platform.

Talks

Unifying Random Fourier Features and Leverage Score Sampling for Kernel Matrix Approximation

Microsoft Research, Redmond

Sept. 2017

Sketching and Sampling Methods for Large Scale Linear Algebra

Twitter, Cambridge

July 2017

Recursive Sampling for the Nyström Method

New England Machine Learning Day Poster Session

May 2017

Algorithms for Determining Tournament Payout Structures

Meeting on Algorithm Engineering and Experiments (ALENEX)	<i>Jan. 2017</i>
Introduction to Linear Sketching	
MIT, Theory Retreat	<i>Sept. 2016</i>
Principal Component Projection without Principal Component Analysis	
International Conference on Machine Learning (ICML)	<i>June 2016</i>
Ridge Leverage Score Sampling	
National Institute of Informatics, Shonan Meeting	<i>July 2016</i>
MIT, Algorithms & Complexity Seminar	<i>July 2016</i>
University of Utah, Data Group Meeting	<i>Jan. 2016</i>
Randomized Block Krylov Methods	
Neural Information Processing Systems (NIPS) Poster Session	<i>Dec. 2015</i>
Large Scale Column Subset Selection	
Yahoo Labs, Science Week Poster Session	<i>July 2015</i>
Dimensionality Reduction for K-Means Clustering	
IBM T.J. Watson, Mathematical Sciences Research Seminar	<i>Aug. 2015</i>
Symposium on Theory of Computing (STOC)	<i>June 2015</i>
MIT, Theory Lunch	<i>June 2014</i>
Principled Sampling for Anomaly Detection	
MIT, Defense Advanced Research Projects Agency Site Visit	<i>Mar. 2015</i>
Network and Distributed System Security Symposium (NDSS)	<i>Feb. 2015</i>
Uniform Sampling for Matrix Approximation	
Yahoo Labs, Scalable Machine Learning Research Seminar	<i>June 2015</i>
Single Pass Spectral Sparsification in Dynamic Streams	
MIT, Annual Sublinear Algorithms Day Poster Session	<i>Apr. 2015</i>
Harvard University, Theory Seminar	<i>Nov. 2014</i>
Foundations of Computer Science (FOCS)	<i>Oct. 2014</i>

Service

Conference Reviewer.....
COLT (2016), ESA (2015, 2017), FOCS (2015), ICALP (2015, 2017), ITCS (2018), NIPS (2015, 2016, 2017), NDSS (2015), RANDOM (2017), SODA (2016, 2017, 2018)

Journal Reviewer.....
IEEE Transactions on Knowledge and Data Engineering (2017), IEEE Transactions on Signal Processing (2016, 2017, 2018), SIAM Journal on Matrix Analysis and Applications (2016, 2017)

Other.....

CSAIL Algorithms Office Hours

Member *2016 – present*
Meet with researchers, generally in applied fields, who seek advice on solving or framing algorithmic problems that appear in their research.

Other Research Experience

Japan National Institute of Informatics

2016

Invited participant, Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra

Honors and Awards

National Science Foundation

2014-2017

Graduate Research Fellowship, recipient.

Yale University

2012

Cum Laude, Distinction in Both Majors.

Professional Experience

Redfin

Seattle, WA

Software Engineer, Data Team

2012 – 2014

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.

Elysium Digital

Cambridge, MA

Summer Consultant

Summer 2011

Provided technical expertise in software patent cases at leading litigation consulting company.

Amicus

New Haven, CT

Software Developer

2010 – 2011

Built tools for political campaign management and social fundraising at an early-stage startup that went on to raise nearly 4 million dollars in seed funding.

Moody's Investors Service

New York, NY

Intern, Asset Finance Group

Summer 2009

Established and monitored bond ratings on securitizations in three asset classes. Wrote code to automate analysis for student loan data and to compile records on aircraft leases.

U.S. Department of Agriculture

East Greenwich, RI

Intern, Natural Resources Conservation Service

Summer 2007

Collected and analyzed data for two major state conservation and research projects. Integrated mapping software and database tools to develop a system for compiling and presenting soil data.